What Determines the Effectiveness of 3D Field of Junctions?

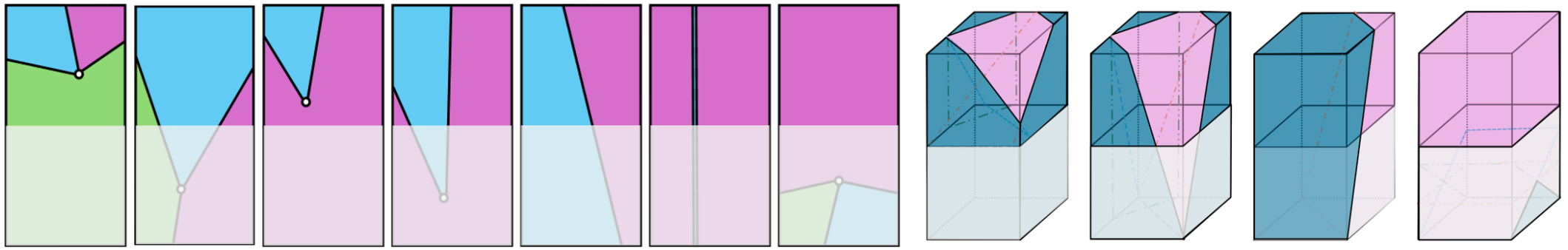
A Study of Volumetric Denoising Applications

SECTION C · TEAM 17

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github.com/samudraneel05/SectionC_Team17

3D Field of Junctions



Junction schematic from the original paper (Moeini et al., ECCV 2026)

Training-free

No network, no dataset, no training. The model is fit directly on the single noisy input volume.

Wedges per patch

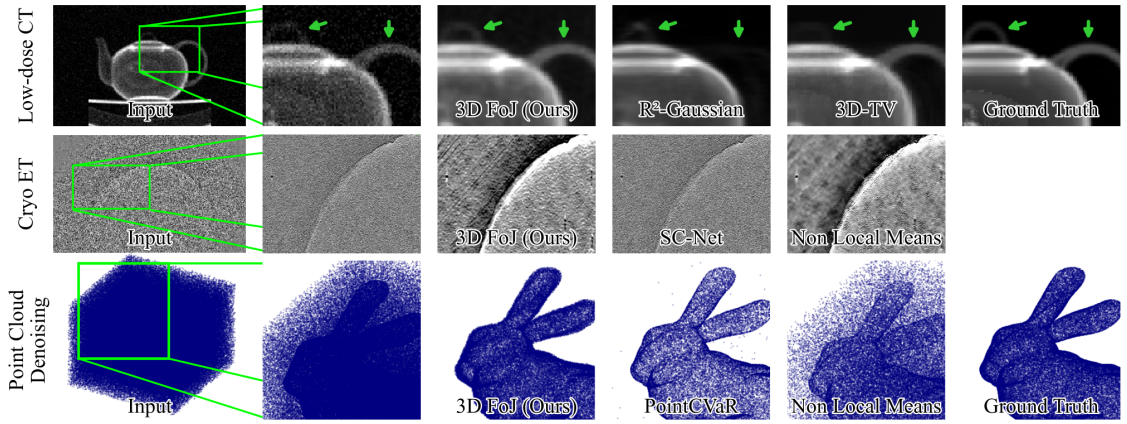
Every R^3 cubic patch is split by planes through one shared vertex into wedge regions; each region takes the mean of the voxels it covers.

Overlap averaging

Patches tile the volume on a grid. Each voxel is rendered by several patches and the outputs are averaged into the final volume.

Moeini, Kim, Romberg & Fridovich-Keil — released implementation used as-is

Applications of 3D Field of Junctions



Original teaser: low-dose CT, cryo-ET, point clouds

- Demonstrated on sparse-view CT, cryo-electron tomography, and point-cloud denoising.
- Open question: does the wedge prior transfer to volumes with different structure?
- We apply the released implementation, unchanged, to three new domains.

Bentheim sandstone micro-CT

Grains and pores separated by sharp interfaces, 10–35 voxels across — exactly the shape the wedge prior assumes.

BrainWeb T1 MRI phantom

Anatomical tissue has smooth gradients, not plateaus - tests the piecewise-flat assumption.

Binary porous media

Sphere pack + Berea scan with ~2-voxel features, smaller than any patch - tests the scale assumption.

Three requirements for effectiveness

The nominal model

A coupled field of junctions: boundary- and color-consistency terms tie overlapping patches together, weighted by λ ramps over refinement iterations.

What runs at the released defaults

With N_ref = 1 the consistency weights never engage, and the single refinement pass renders from pre-update parameters. Effectively: an independent wedge fit per patch + overlap averaging.

Three requirements for FoJ to help:

1 · Unbiased input

Region intensities are voxel means. A systematic offset in the input is reproduced, not removed e.g. clamping a CGLS reconstruction passes straight through.

2 · Structure ≥ patch scale

Wedges are piecewise-constant. Geometry finer than the patch, or smooth gradients, cannot be expressed as the fit collapses toward the mean.

3 · Noise above crossover

The wedge model is biased. It wins only when the noise it averages away costs more than the bias it introduces.

Step 0 - Reproduce the published benchmark

- Sparse-view CT ablation: engine volume, 20 cone-beam views, P50 Poisson noise.
- Pipeline rebuilt end-to-end: R²-Gaussian benchmark + TIGRE projector + 60 CGLS iterations, then the released FoJ code.
- Qualitative result reproduces exactly: quality rises monotonically with patch size R, best at R = 12, with shrinking marginal gains.
- Absolute PSNR sits +1.2 to +3.4 dB above reported — consistent with the released M = 2 (vs. M = 3 in the paper) and unspecified reconstruction details.

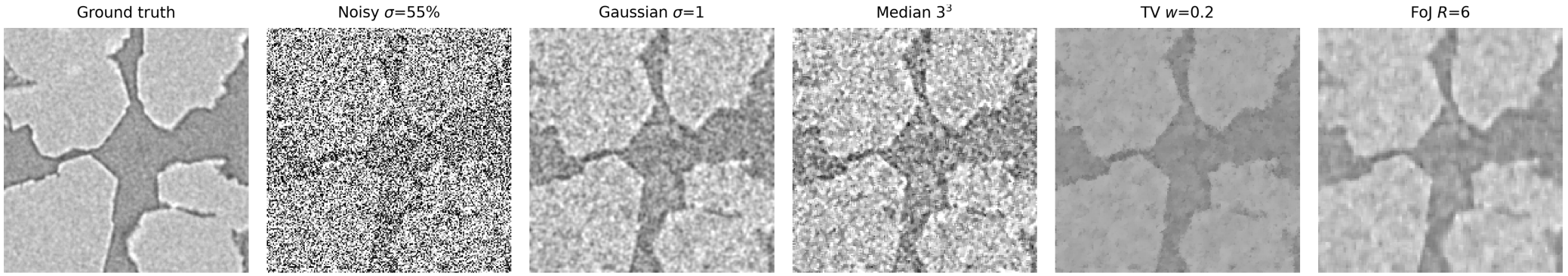
R	Reported	Ours	Δ (dB)
4	11.33	14.72	**+3.39**
6	14.37	16.93	**+2.56**
8	16.18	17.99	**+1.81**
10	17.17	18.58	**+1.41**
12	17.70	18.86	**+1.16**

Engine-volume CT ablation — denoised PSNR vs. patch size R

+1.2 → +3.4 dB

ours above reported, gap narrows as R grows

Bentheim micro-CT: where FoJ wins



σ = 55% noise -> FoJ keeps grain–pore interfaces sharp while Gaussian, median, and TV blur

σ	Noisy	Best classical	FoJ
15%	16.89	**28.51**	26.55
30%	12.00	**26.03**	25.64
40%	10.28	23.39	**24.93**
55%	8.76	21.05	**24.29**

3D PSNR (dB), 192³ crop of Bentheim sandstone

+3.24 dB

FoJ over the best classical filter at σ = 55%

≈ 11 dB

input-PSNR crossover — below this, the wedge prior pays off; above it, classical filters win

BrainWeb MRI: where FoJ fails due to smooth texture

- BrainWeb T1 phantom (181×217×181), Rician noise σ = 3–13%.
- Tissue gradients are smooth, so constant wedges quantize them into steps — model bias at every noise level.
- FoJ trails the 3^3 median filter at all five levels; larger patches only make it worse.
- Same story in reconstruction: as a plug-and-play denoiser, FoJ-PnP reaches 27.99 dB vs. 30.39 dB for TV-PnP at 4× undersampling.

σ	Median	FoJ R=6	gap
3%	30.80	25.67	**−5.13**
7%	24.58	22.52	**−2.06**
13%	19.28	18.31	**−0.97**

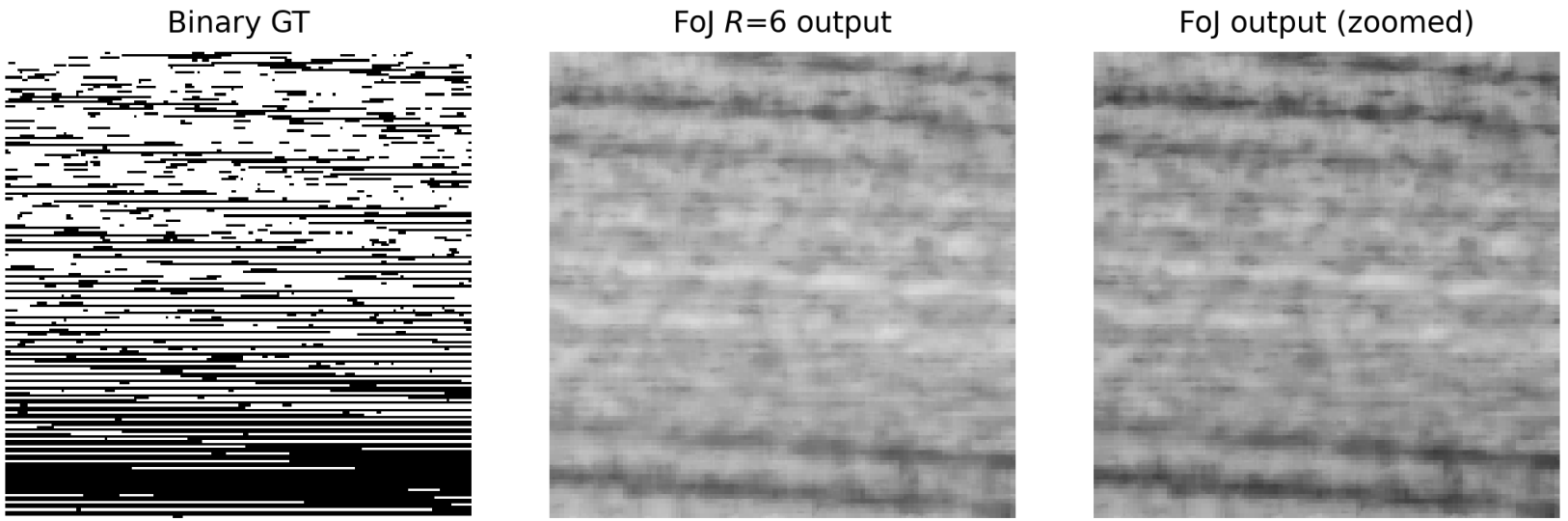
−5.13 dB

FoJ R=6 vs. median filter at σ = 3% — below the noisy input itself

0 / 5

noise levels where FoJ beats any classical baseline on MRI

Binary porous media: where FoJ fails due to sub-patch features



Berea, σ = 15%: binary ground truth (left) → FoJ output (right) collapses toward the volume mean

19.5 → 7.1 dB

FoJ scores below the noisy input — denoising destroys the signal it should keep.

σ 0.49 → 0.09

Output intensity spread shrinks 5×: the volume comes out nearly flat.

Why

Features are ~2 voxels; three planes through one vertex can't isolate dozens of pores in an R^3 patch, so every wedge mean ≈ global mean.

Explanation of all three outcomes

FoJ helps only when all three hold:

- unbiased input (means reproduce offsets),
- piecewise-flat structure at ≥ patch scale,
- noise high enough to pay for the wedge bias.

✓ Bentheim micro-CT

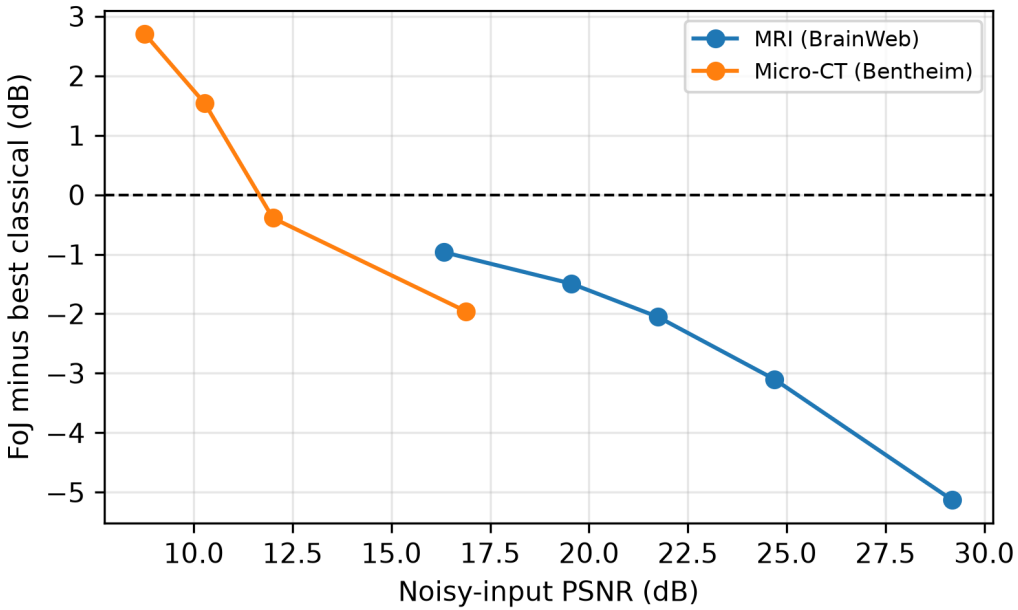
piecewise-flat grains > patch + input < 11 dB → best method (+3.2 dB)

✗ BrainWeb MRI

smooth gradients violate flatness → loses at every noise level

✗ Binary porous media

~2-voxel features < patch → collapses to the mean



FoJ margin (PSNR − best classical) vs. input PSNR across domains

Takeaways

3D FoJ is a structural prior, not a universal denoiser.

It wins where structure is piecewise-flat at patch scale and noise is severe like Bentheim micro-CT at 8.8 dB input: +3.2 dB over every classical filter.

It loses where gradients are smooth (MRI) or features are finer than a patch (binary media). Classical filters are better and cheaper.

Before adopting it check three things: is the input unbiased, is the structure expressible at patch scale, and is the noise above the crossover.

github.com/samudraneel05/SectionC_Team17

Thank you